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Continuous Assessment 1 (CA 1) - Questions

Course Code: CSE563	Course Title: APPLIED MATHEMATICS
Section(s): K3P25RB	Syllabus Covered: Unit 1 and Unit 2
Total Marks: 30 Time Allowed: 60 minutes	Maps to CO1 and CO2 - basic mathematical concepts; vectors and matrices

Unit I: Basic Mathematics in Computer Science · Unit II: Vectors and Matrices

Instructions: attempt all questions. Show every step of working clearly; an unsupported final answer receives no marks. The marks for each question are shown in brackets.

Part A - Short Answer Questions (6 Marks - 3 questions x 2 marks)

1. Compute gcd(360, 132) using the Euclidean algorithm, and hence find integers x, y such that 360x + 132y = gcd(360, 132) using the extended Euclidean algorithm. **(2 Marks)**

2. For the vectors u = (2, −1, 3) and v = (1, 4, −2) in $\mathbb{R}^3$, compute: (a) the dot product u · v; (b) the Euclidean norm ‖u‖; and (c) the orthogonal vector projection proj_u v. **(2 Marks)**

3. In an image resampling pipeline, the four corner pixel intensities on the unit square grid are f(0,0) = 15, f(1,0) = 25, f(0,1) = 35 and f(1,1) = 65. Calculate the interpolated intensity at the point (0.4, 0.6) using bilinear interpolation, and state the value obtained by nearest-neighbour interpolation. **(2 Marks)**

Part B - Core Numerical Problems (12 Marks - 2 questions x 6 marks)

4. Consider the 4-variable Boolean switching function F(A, B, C, D) = Σm(0, 2, 5, 7, 8, 10, 13, 15). **(6 Marks)**

(a) Plot the function on a 4 × 4 Karnaugh map with Gray-code ordering (CD, AB = 00, 01, 11, 10).

(b) Identify all prime implicants, circle the maximal groupings on the map, and write the minimal Sum-of-Products (SOP) expression.

(c) Algebraically verify that your minimal expression produces the correct output for minterms m_{10} (1010) and m_7 (0111), and rejects the non-minterm m_1 (0001).

5. Given the real symmetric matrix A = [[5, 2], [2, 2]]: **(6 Marks)**

(a) Find the characteristic polynomial and calculate all eigenvalues λ_1, λ_2.

(b) Determine the corresponding eigenvectors v_1, v_2, normalise them to unit length, and verify that their dot product is 0.

(c) Construct the orthogonal modal matrix P = [$\hat{v}_1$ $\hat{v}_2$] and confirm that $P^TAP = D = \text{diag}(\lambda_1, \lambda_2)$.

Part C - Applied and Analytical Problems (12 Marks - 2 questions x 6 marks)

6. Calculus optimisation and mathematical induction: **(6 Marks)**

(a) A loss surface in machine learning is modelled by $f(x, y) = 2x^2 + 4xy + 3y^2 - 6x - 8y + 5$. (i) Find the gradient vector ∇f(x, y). (ii) Determine the stationary point by setting ∇f = (0, 0). (iii) Taking the initial point (0, 0) and learning rate α = 0.1, compute the updated point after one gradient-descent step, and verify that the function value decreased.

(b) Prove by mathematical induction that $\Sigma\,(2i - 1)$ from i = 1 to n equals n^2 for every integer $n \geq 1$. Explicitly state the base case, the induction hypothesis and the inductive step.

7. Dimensionality reduction and composite image transformations: **(6 Marks)**

(a) A 2-D feature dataset has sample covariance matrix Σ = [[6, 2], [2, 3]]. (i) Find the eigenvalues λ_1, λ_2 of Σ. (ii) Compute the total variance and the percentage of variance explained by PC1. (iii) Compute the unit eigenvector $\hat{e}_1$ corresponding to PC1.

(b) A digital 2-D object point P(2, 3) is transformed using homogeneous coordinates. (i) Write the 3 × 3 matrix for a counter-clockwise rotation by 90° about the origin. (ii) Write the 3 × 3 matrix for a translation by t = (4, −1). (iii) Compute the composite matrix M = M_trans M_rot and find the final coordinates of P′.

----- End of Question Paper -----